

Mixing and Filling Operations using PLC, SCADA and IOT

Dr. K. Vadivukkarasi

Department of Electronics and
Communication
SRM Institute of Science and
Technology
Chennai, India
vadivukk@srmist.edu.in

Shreyans Bhargava

Department of Electronics and
Communication
SRM Institute of Science and
Technology
Jodhpur, India
sb1628@srmist.edu.in

Abstract— in this paper, an automated liquid mixing and filling that imitates a real system is developed, utilizing Delta Programmable Logic Controller (PLC) DVP14SS2 as the Master controller programmed by using WPLSoft 2.51 software. However, the programming language motivated by international standard IEC 61131-3. System monitored using DOPSoft 4.00.16- Supervisor software As the Supervisory Control and Data Acquisition (SCADA) System. IOT Module ESP8266EX is used for system start & stop remotely. The software provides a user friendly Human Machine Interface (HMI) to monitor and control this system. Ladder diagrams are used to design the system's operating sequence. The components used in this work are programmable logic controller (PLC), Electromagnetic relays in PLC, proximity sensor, push buttons for start & emergency stop, level switches, solenoid valves, and DC Motor, which consists of three sections such as the conveyor section, filling section and mixing section. This work will be useful in paint industries, concrete industries, Beverage processing industry and Food processing technology.

Keywords—SCADA, PLC, Ladder Logic, IOT.

I. INTRODUCTION

To withstand the fierce competition, Companies everywhere need to use the newest technology, including automation, to survive the intense competition. Industries have greatly benefited from the automation industry. In the process of producing goods and services, automation is a technology used to decrease human contact, speed up processes, and enhance quality. Optimizing efficiency is crucial in today's competitive manufacturing industry. This project meets that demand by integrating Supervisory Control and Data Acquisition (SCADA) with Programmable Logic Controllers (PLC) in a smooth manner. The project's core is PLC technology, which regulates and automates the complex mixing and filling processes on a massive scale. Its ability to make decisions in real time assures precision, dependability, and productivity in the manufacturing process. The SCADA system offers a centralized monitoring and control interface, which improves PLC capability. This interface gives operators complete control over the system, allowing them to monitor and modify parameters remotely, track production data in real time, and react quickly to any abnormalities.

In addition to enhancing efficiency in operation, PLC and SCADA work together to decrease human error, minimize downtime, and increase product quality. The project's comprehensive approach to automation has the potential to reinvent production standards and promote an industrial ecosystem that is more adaptable and competitive. The production environment can become more adaptable and expandable with the help of automation. The system is a versatile solution that fulfills the changing demands of today's manufacturing sector since it is simple to modify to meet changing production demands. This flexibility enables proactive problem-solving, maintenance planning, and well-informed decision-making, especially when combined with the capacity for precise information collecting and analysis. In day to day life industries have to handle many challenges like rapid increase in the productivity and economic expenditure even though industries have to function safely. However, with earlier systems, if production increases steadily, maintenance costs would also rise. With addition, these systems have a number of drawbacks, including high costs, inactivity, and dangerous operation. Automation can be employed as a solution to these issues. In simple terms, this effort is a major step toward an automated industrial future where intelligent systems work together to improve productivity, simplify procedures, and raise the overall efficiency of large-scale mixing and filling processes. Industries may expect not only immediate production gains from this integration, but also for a long-time sustainable growth and innovation. Use of advanced technology for automation, the Large-Scale Mixing and Filling Operation through the PLC and SCADA Integration project is a unique method for improving manufacturing operations. Optimizing efficiency is crucial in today's competitive manufacturing industry. This project meets that demand by integrating Supervisory Control and Data Acquisition (SCADA) with Programmable Logic Controllers (PLC) in a smooth manner. The project's core is PLC technology, which regulates and automates the complex mixing and filling processes on a massive scale. Its ability to make decisions in real time assures precision, dependability, and productivity in the manufacturing process. The SCADA system offers a centralized monitoring and control interface, which improves PLC capability. This interface gives operators complete control over the system, allowing them

to monitor and modify parameters remotely, track production data in real time, and react quickly to any abnormalities. In addition to enhancing efficiency in operation, PLC and SCADA work together to decrease human error, minimize downtime, and increase product quality. The project's comprehensive approach to automation has the potential to reinvent production standards and promote an industrial ecosystem that is more adaptable and competitive. By adopting with this innovative technical integration, organizations could look forward to increased output, reduced costs, and a more efficient production process, which will eventually result in significant improvements in the field of large-scale mixing and filling processes.

The production environment can become more adaptable and expandable with the combination of PLC and SCADA. The system is a versatile solution that fulfills the changing demands of today's manufacturing sector since it is simple to modify to meet changing production demands. This flexibility, along with the ability for real-time data collection and analysis, allows proactive problem-solving, maintenance planning, and informed decision-making. In simple terms, this effort is a major step toward an automated industrial future where intelligent systems work together to improve productivity, simplify procedures, and raise the overall efficiency of large-scale mixing and filling processes. Industries may expect not only immediate production gains from this integration, but also for a long-time sustainable growth and innovation.

II. DEVICE STRUCTURE AND MODELING

In this project, the PLC is the system's brain. PLC is included into the project's device structure to regulate mixing and filling operations in real-time according to the program uploaded to it. The PLC is synchronized with SCADA systems to provide centralized monitoring. In this work there are three sections such as the conveyor section, filling section and mixing section are carried out controlled by the Delta PLC with 8 Digital Inputs and 6 Digital Outputs that complies with the IEC 61131-3. The SMPS, switched mode power supply, transforms 230 V of input into 24–27 V of output. Push button switches that are used to turn on or off supplies. They are powered by tiny electric currents or ON/OFF command can be given by IOT module also two 24 volt ½" size valve specification solenoid valve are given to two different tanks. Operation of the solenoid valves is set by the level switches which are installed on mixing tank. A 24V 30 RPM DC motor is utilized for liquid mixing. Filling of the bottle depending on the mixing tank level switch feedback and it stopping is set by timer. A conveyor belt is used to carry the final mixed liquid filled bottle which is operated by a 24V 200W 20 RPM DC motor. A 24V DC, Detection distance 8 mm rated capacitive type proximity sensor is mounted on the conveyor belt for move away the filled bottle & bring in an empty one. Figure-1 shows the schematic diagram of liquid mixing and filling process. The process consists of four tanks, three tanks (Tank-1, Tank-1, and Mixer Tank) two different tanks for different liquid and in tank is used for liquid mixing. Conveyor belt is used for the movement of bottle. Two solenoid valves are used for filling the mixing tank and one solenoid valve is used for bottle filling.

Delta PLC is used to control the entire mixing and filling process by ladder logic.

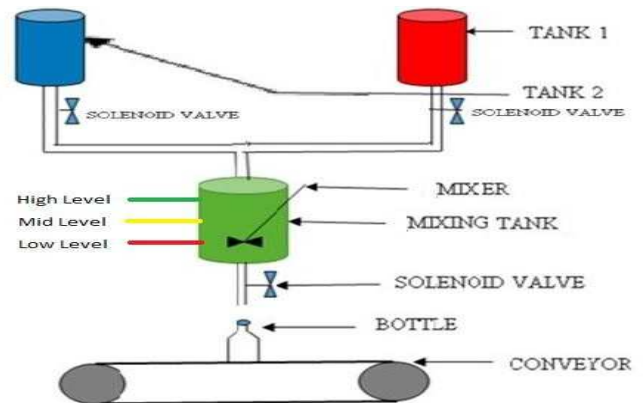


Fig.1. Schematic diagram of liquid mixing and filling

III. SIMULATION AND SYSTEM DESIGN

A. Programming Logic

Figure-2 represents the ladder diagram of Mixer tank level checking. Figure-3 shows the ladder logic of mixer tank filling process by Tank-1 & Tank-2. Figure-4 & Figure-5 expresses the logic of liquid mixing and bottle filling respectively.

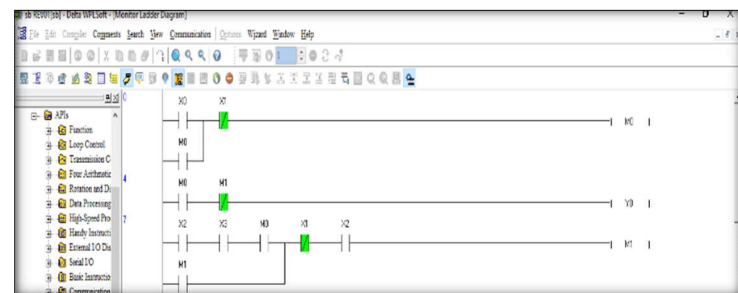


Fig. 2 Ladder Dia. For Level checking of liquids



Fig. 3 Ladder Dia. For liquids filling

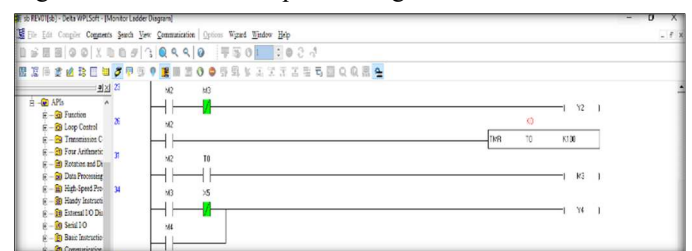


Fig. 4 Ladder Dia for Liquid mixing

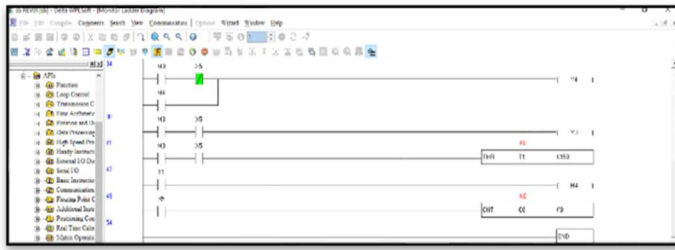


Fig. 5 Ladder Dia. For mixed liquids discharge to bottle with belt conveyor.

B. System Block Diagram

In industrial automation, PLC (Programmable Logic Controller) is often used as a central control unit for machinery and processes. It's not uncommon for PLCs to communicate with other devices, including PCs, to exchange data, monitor processes, or provide control interfaces. Engineering PC into a PLC system can be done for various reasons, such as: Data Logging and Analysis, User Interface, Networking and Communication, Remote Monitoring and Control, PLC Programme Development. In the hardware architecture of SCADA systems, the Engineering PC shares the data between PLC/ SCADA software and PLC Hardware unit through Ethernet. WPLSoft 2.51 software for PLC Programming and DOPSoft 4.00.16-Supervisor software for SCADA system. Integrating an IoT (Internet of Things) module ESP8266EX into a PLC system can enhance its capabilities by enabling remote monitoring, control, and data analysis through internet connectivity. The block diagram of System structure is shown in Fig. 6

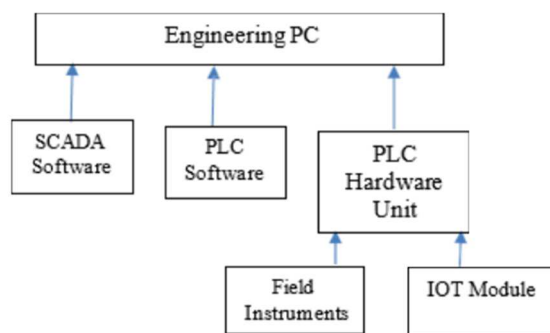


Fig. 6 Block Diagram of the System.

C. Process & Flow Chart

The flow diagram for the liquid mixing and filling operation is shown in **Figure 1**. The liquid mixing and filling procedure involves the following steps.

1. Process Start.
2. Level will be checked in filling tanks A & B.
3. If Level Switch 1 (Tank-A) & Level Switch 2 (Tank-B) of Tank A & B are high then step 4. Else go to step (2).

4. Levels will be checked in mixing tank by Level Switch -3, 4 & 5. (Level Switch-3 for Low Level, Level Switch-4 for Mid-Level & Level Switch-5 for High Level.)
5. If Level in mixing tank is below Low Level go to STEP-8. Else go to step STEP-6.
6. If Level in mixing tank is above Low Level (LS- 3 High) but below middle level (LS-4 Low) go to STEP- 8. Else go to step STEP-7.
7. If Level in mixing tank is above middle Level (LS-3 & LS-4 High) but below High level (LS- 5 Low) go to STEP- 9. Else go to step STEP-8.
8. Solenoid Valve below Tank-A will get on allowing discharge of material in mixing tank, till the middle level is achieved. LS-3 & 4 (Low & middle Level Switch) gets high. Post that solenoid valve-A will get off.
9. Solenoid Valve below Tank-B will get on allowing discharge of material in mixing tank, till the high level is achieved. LS-3, 4 & 5 (Low, middle & High Level Switch) gets high. Post that solenoid valve-B will get off.
10. Mixer Motor will get ON for the pre-set time stetted via PLC timer. The Motor On time can be adjusted as per convince. After timer set time the motor will stop automatically.
11. Conveyor Belt will start carrying the bottle that needs to be filled to the discharge end of mixing tank.
12. Proximity Switch Installed on Conveyor Belt will sense the bottle when the bottle comes to the discharge end of mixing tank & it will stop Conveyor Belt
13. Solenoid Valve-C below Mixing Tank will get on for the pre-set time stetted via PLC timer allowing discharge of material into the bottle. The Solenoid Valve On time can be adjusted as per convince. After timer set time the Solenoid Valve will stop automatically & Conveyor Belt will start.
14. The Same process is repeated till Tank A & B have sufficient Levels.

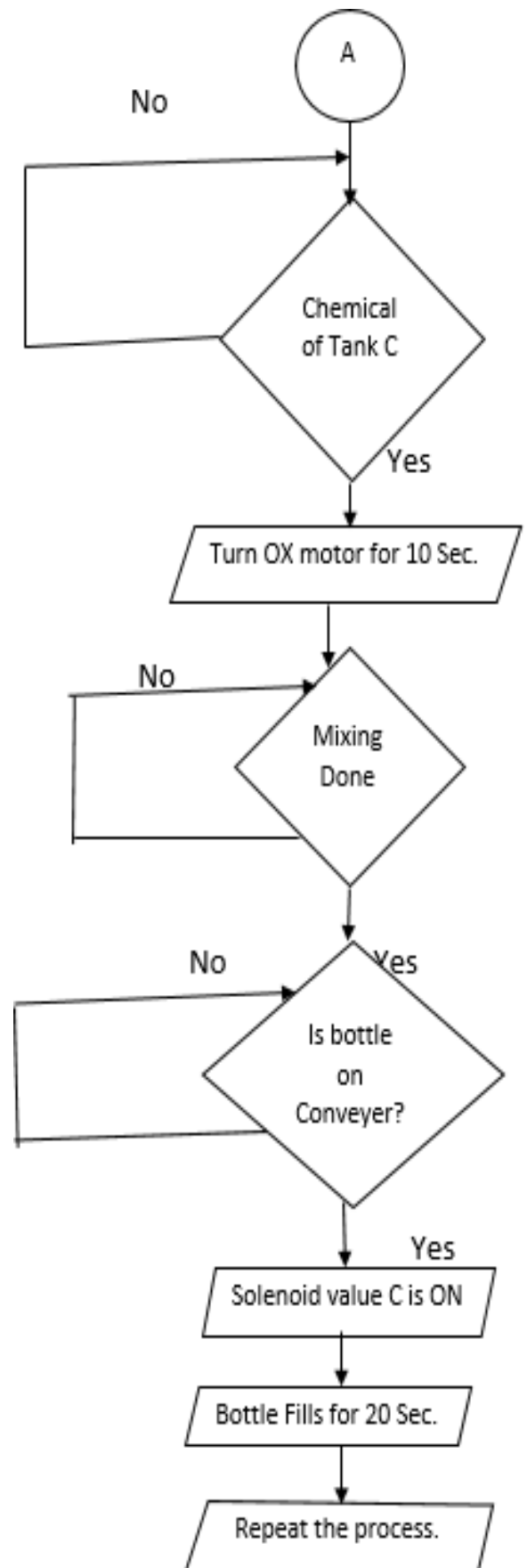
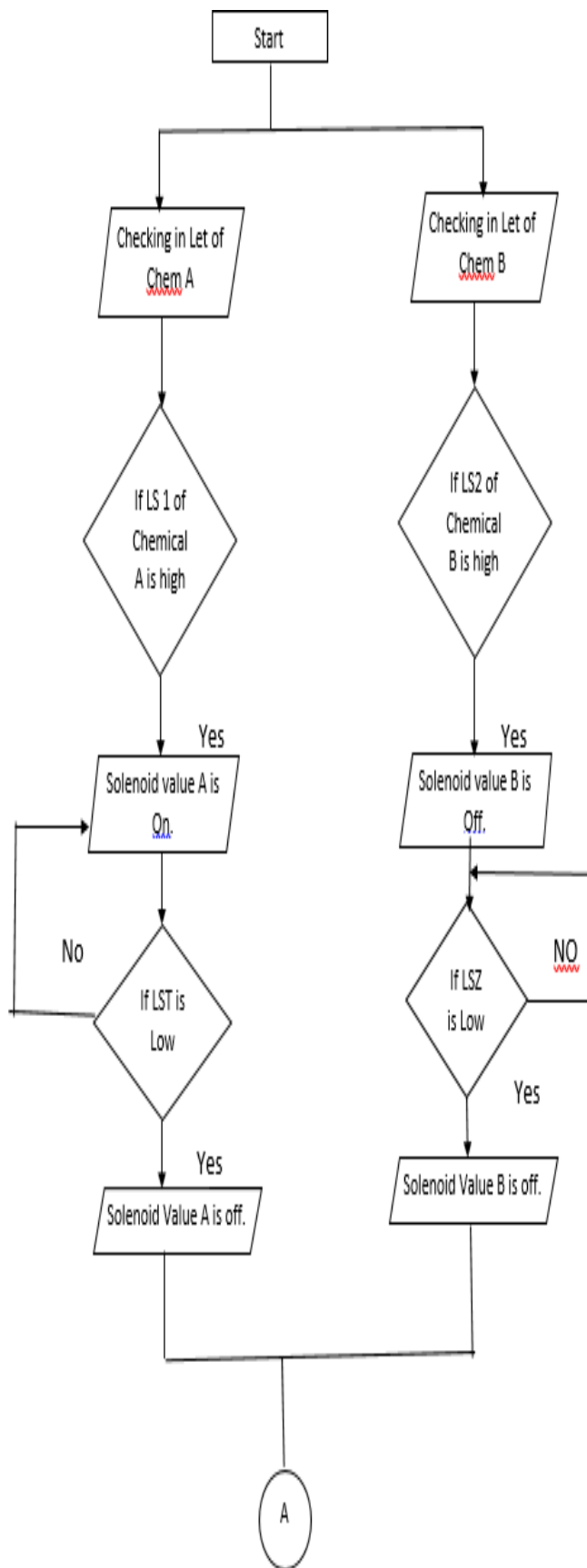


Fig- 6 Flow Chart of the System

IV. RESULTS

PLC-assisted mixing and filling produces the correct composition by precisely filling bottles with liquid. The procedure reduces the amount of liquid waste to zero percent. The adoption of PLC improved efficiency (Fig.7 shows the efficiency and speed of this system) because, while a manual operation can take one to two minutes, a PLC can finish the same task in 20 seconds. When the PLC and IOT (Internet of Things) are combined, the operator can quickly access starting and stopping operations. The emergency switch connected to the logic allows the operator to stop the plant in the event of a hazard like equipment failure. Emergency stop system is used for human safety as well as protecting the field devices .

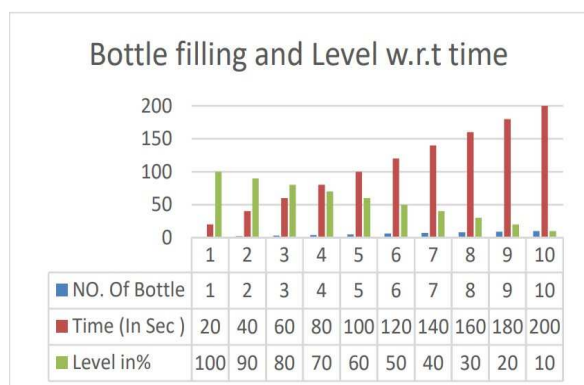


Fig 7. Bottle filling time and level.

^a. Sample of a Tablefootnote. (Table footnote)

Fig. 1. Example of a figure caption. (figure caption)

Figure Labels: Use 8 point Times New Roman for Figure labels. Use words rather than symbols or abbreviations when writing Figure axis labels to avoid confusing the reader. As an example, write the quantity “Magnetization”, or “Magnetization, M”, not just “M”. If including units in the label, present them within parentheses. Do not label axes only with units. In the example, write “Magnetization (A/m)” or “Magnetization {A[m(1)]}”, not just “A/m”. Do not label axes with a ratio of quantities and units. For example, write “Temperature (K)”, not “Temperature/K”.

V. CONCLUSION

A revolutionary step in industrial automation, the Large-Scale Mixing and Filling Operation through the PLC and SCADA Integration project precisely and effectively handles the complexities of large-scale manufacturing. Using the smooth integration of SCADA's centralized monitoring and PLC technology's real-time control, the project improves processes, reducing human error, boosting product quality, and eventually increasing productivity. It has been demonstrated that the suggested technique works well in preventing needless liquid spills and waste to 0%. The system's flexibility provides scalability and response to increasing production demands, providing the foundation for an innovative and competitive manufacturing environment. Industries that adopt this advanced integration can expect to see improvements in productivity right away as well as a solid basis for future development, innovation, and competition in the dynamic field of automation in industry. In liquid mixing and filling processes, accuracy is crucial to ensure that the desired proportions of ingredients

are mixed correctly, and products are filled precisely to the specified levels. Accuracy and consistency in manual processes heavily rely on the skill and attention of the operators. Variability is inherent due to human factors such as fatigue, distractions, and differences in technique. Automated processes are inherently more accurate and consistent since they rely on precise control algorithms and calibrated equipment. Automated systems can maintain consistent mixing ratios and fill volumes batch after batch, resulting in higher product quality and repeatability. With this method we can reduce human error to 0% and achieve accuracy over 90%. Achieving accuracy in these processes involves a combination of precise measurement, control, and monitoring techniques. Here's how industrial automation ensures accuracy in liquid mixing and filling processes: The project provides example of how innovative technologies can be used to transform traditional industrial concepts in order to create a more intelligent and profitable future.

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